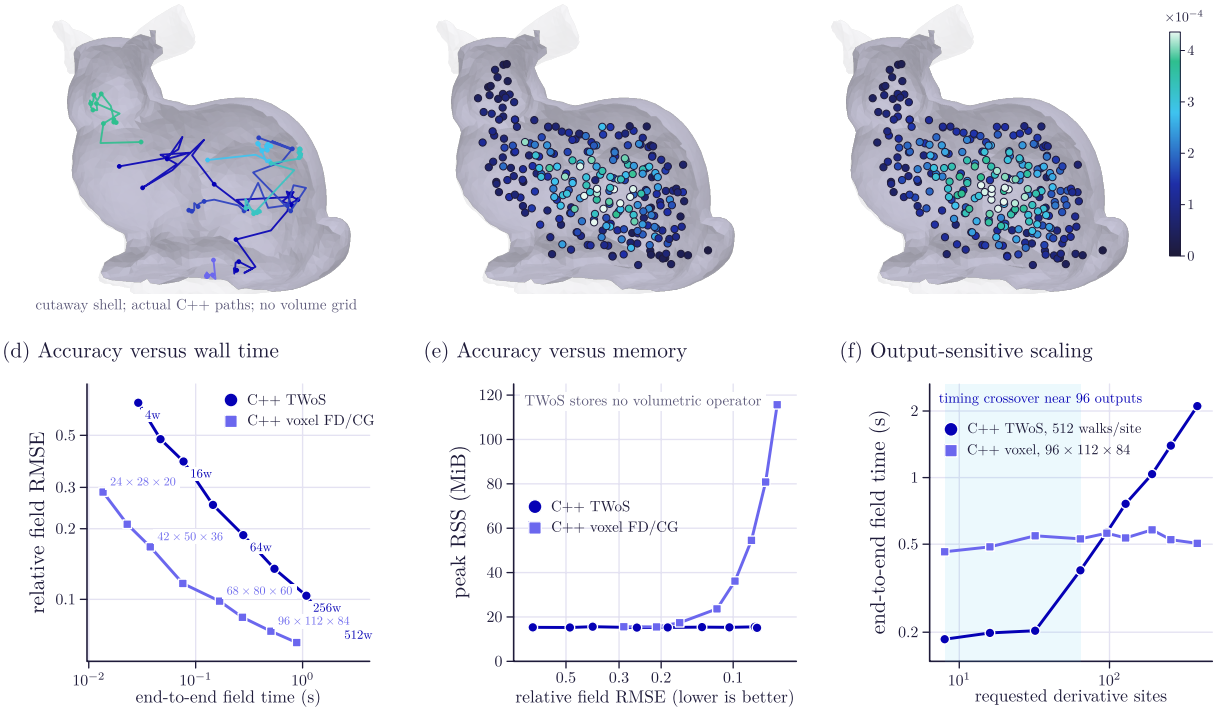

(a) Embree-accelerated sphere walks
(b) Direct TWoS score $-D_T$
(c) Voxel-adjoint score $-D_T$
$\times 10^{-4}$
4
3
2
1
0
cutaway shell; actual C++ paths; no volume grid
(d) Accuracy versus wall time
C++ TWoS
C++ voxel FD/CG
4w
16w
64w
256w
512w
24 × 28 × 20
42 × 50 × 36
68 × 80 × 60
96 × 112 × 84
relative field RMSE
0.5
0.3
0.2
0.1
10^{-2}
10^{-1}
10^{0}
end-to-end field time (s)
(e) Accuracy versus memory
TWoS stores no volumetric operator
C++ TWoS
C++ voxel FD/CG
peak RSS (MiB)
120
100
80
60
40
20
0
0.5
0.3
0.2
0.1
relative field RMSE (lower is better)
(f) Output-sensitive scaling
timing crossover near 96 outputs
C++ TWoS, 512 walks/site
C++ voxel, 96 × 112 × 84
end-to-end field time (s)
2
1
0.5
0.2
10^{1}
10^{2}
requested derivative sites